

Milena Rmus

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EDUCATION

University of California, Berkeley

PhD, Cognitive Science

Aug. 2019 – May 2024

Berkeley, CA

Brown University

BS, Cognitive Neuroscience (Magna Cum Laude)

Aug. 2014 – May 2018

Providence, RI

EXPERIENCE

Roundtable AI

Member of Technical Staff

May 2024 – present

San Francisco, CA

- Member of the founding engineering team designing and **training machine learning models (gradient boosting)** for distinguishing human and agentic behavior online, **improving the detection accuracy to 99%**.
- Contributed to the companys research direction in AI safety and evaluation, including work submitted to leading machine learning conferences (NeurIPS, 2026).
- **Supported approximately 3x customer growth within the companys first six months** while simultaneously scaling machine learning infrastructure and research workflows.
- **Contributed to frontend engineering and dashboard design for the Roundtable web application**, improving usability and experiment monitoring workflows .

Helmholtz Institute for Human-Centered AI

Research Scientist

May 2024 – present

Munich, Germany

- Led a research project on **reverse-engineering interpretable algorithms from behavioral datasets using Large Language Models (LLMs; Llama 3, Qwen 2.5, R1)**, with applications in human decision modeling and behavioral model discovery.
- Built model-fitting and evaluation tools and workflows using **Python, Hugging Face Transformers, PyTorch, and SciPy to support automated scientific discovery**.
- Developed benchmarking workflows with **Bayesian model selection, automated error checking, and iterative model revision via natural language feedback loops**.
- Supervised a cross-disciplinary team of junior researchers and collaborated with ML engineers to translate cognitive modeling insights into practical, scalable systems.

Lawrence Livermore National Laboratory

Data Science Intern

May 2022 – Aug. 2022

Livermore, CA

- Optimized amino acid sequences as mathematical expressions using Pareto optimization in Deep Symbolic Regression.
- **Achieved a 2+ term reduction in model complexity while preserving performance.**
- Built random forest classifiers (AUC = 0.88) to predict compound binding affinity from molecular descriptors.

UC Berkeley

Graduate Student Instructor (Computational Models of Cognition)

Sep. 2020 – Dec. 2020; Sep. 2023 – Dec. 2023

Berkeley, CA

- Designed and delivered discussion materials on algorithmic and neural architecture parallels between cognitive science and artificial intelligence.
- Synthesized advanced course content into accessible formats for diverse student backgrounds.
- Set up and maintained course infrastructure in collaboration with UC Berkeley Data Lab, deploying Jupyter Notebook servers for interactive, reproducible problem sets.

Princeton University

Research Specialist

Jun. 2018 – Jun. 2019

Princeton, NJ

- Oversaw experiment rollout, cross-site data integrity, and ran fMRI scanning sessions.
- **Developed a web app hosted on Amazon Mechanical Turk in JavaScript (jQuery, jsPsych)** to run decision-making experiments, and stored data on Firebase.
- Analyzed behavioral data using machine learning tools (SVMs, PCA, and clustering) in Python.

SELECTED PROJECTS

Using Large Language Models to generate Computational Models of Behavior

- Designed a framework leveraging LLMs (GPT-4, LLama 3, Qwen 2.5, DeepSeek R1) to generate executable Python code for computational cognitive models based on behavioral input data.
- Demonstrated that LLM-generated models **outperform traditional cognitive models in behavioral model fit and parsimony, surpassing literature baselines in 80% of evaluation domains.**
- Developed an automated feedback loop to detect scientific inconsistencies and iteratively refine model hypotheses through natural language interaction and code validation.

Process-Level Evaluation of AI Systems using Cognitive Frameworks (NeurIPS submission)

- Co-developed a battery of 30 cognitive tasks (CogPrint30) designed to distinguish humans from AI agents using process-level behavioral features rather than output performance alone.
- Built machine learning pipelines for large-scale behavioral analysis and classification of human vs. agentic interaction patterns across frontier AI systems.
- Developed evaluation and fine-tuning workflows for large language models using behavioral process features, reinforcement learning environments, and browser-based agent interaction systems.
- Demonstrated that process-level behavioral features reliably distinguish humans from frontier AI systems even when task performance is matched, with classifier performance reaching **AUCs of approximately 0.85–0.88 across tasks.**

Detailed research interests and publications available at: <https://milennacnlab.github.io/MilenaCV.pdf>.

Technical Skills

- **Languages:** Python (expert), MATLAB (expert), JavaScript (fluent), R (fluent), LaTeX (fluent), SQL (prior experience)
- **Libraries & Tools:** Pandas, NumPy, Scikit-Learn, TensorFlow, Keras, PyTorch, Matplotlib, Seaborn, ggplot2, Git, Adobe Illustrator, Procreate